Pattern Recognition and Machine Learning

Chapter 3: Linear Models for Regression

Reading Notes

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Linear Basis Function Models

Simplest Linear Model:

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + \cdots + w_D x_D$$

Linear Model with Basis Functions:

$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\mathbf{x})$$

Vector Form (with $\phi_0(\mathbf{x}) = 1$):

$$y(\mathbf{x}, \mathbf{w}) = \sum_{j=0}^{M-1} w_j \phi_j(\mathbf{x}) = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x})$$

Basis Functions:

- **Polynomial**: $\phi_j(x) = x^j$
- **Gaussian**: $\phi_j(x) = \exp\left(-\frac{(x-\mu_j)^2}{2s^2}\right)$
- **Sigmoidal**: $\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right)$

Maximum Likelihood and Least Squares

Probabilistic Model:

$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \beta^{-1})$$

Likelihood Function:

$$p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}(t_n|\mathbf{w}^T\phi(\mathbf{x}_n), \beta^{-1})$$

Log-Likelihood:

$$\ln p(\mathbf{t}|\mathbf{w}, \beta) = \frac{N}{2}\ln\beta - \frac{N}{2}\ln(2\pi) - \beta E_D(\mathbf{w})$$

Key Equivalence

Maximizing likelihood = Minimizing sum-of-squares error

Maximum Likelihood Solution

$$E_D(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_n)\right)^2$$

$$\nabla_{\mathbf{w}} E_D(\mathbf{w}) = -\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_n)\right)\boldsymbol{\phi}(\mathbf{x}_n)$$

$$\text{Set to zero: } 0 = -\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_n)\right)\boldsymbol{\phi}(\mathbf{x}_n)$$

$$\sum_{n=1}^{N} t_n\boldsymbol{\phi}(\mathbf{x}_n) = \sum_{n=1}^{N}\boldsymbol{\phi}(\mathbf{x}_n)\boldsymbol{\phi}(\mathbf{x}_n)^T\mathbf{w}$$

$$\boldsymbol{\Phi}^T\mathbf{t} = \boldsymbol{\Phi}^T\boldsymbol{\Phi}\mathbf{w}$$

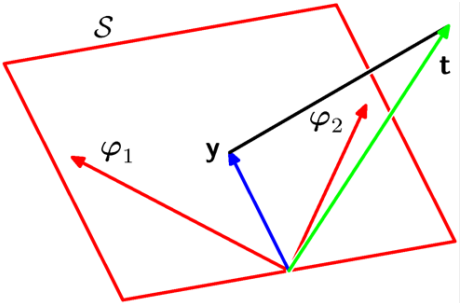
$$\mathbf{w}_{ML} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{t}$$

The design matrix $\boldsymbol{\Phi}$ is defined as:

$$\boldsymbol{\Phi} = \begin{bmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \cdots & \phi_{M-1}(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{bmatrix}$$

- $\phi_0(\mathbf{x}) = 1$ (bias term)
- $\phi_j(\mathbf{x})$: basis functions
- M: number of parameters
- N: number of data points

Geometry of Least Squares



- $\mathbf{t}$: Target vector in N-dimensional space
- φ_j: Basis function vectors
- $\mathbf{y}$: Prediction vector in subspace S

Geometric Meaning of Normal Equations

The condition $\mathbf{\Phi}^T(\mathbf{t} - \mathbf{\Phi w}) = \mathbf{0}$ means the residual vector is orthogonal to all basis functions, ensuring optimal projection onto the model space.

Sequential Learning

Stochastic Gradient Descent:

Parameter Update:

$$\begin{aligned}\mathbf{w}^{(\tau+1)} &= \mathbf{w}^{(\tau)} - \eta \nabla E_n(\mathbf{w}) \\ &= \mathbf{w}^{(\tau)} + \eta(t_n - \mathbf{w}^{(\tau)T}\phi_n)\phi_n\end{aligned}$$

- Also known as **LMS algorithm**
- Processes one data point at a time
- Suitable for large datasets or online learning
- Learning rate η requires careful tuning

Advantage

Avoids storing entire dataset in memory - useful for big data applications

Regularized Least Squares

Regularized Error Function:

$$E(\mathbf{w}) = E_D(\mathbf{w}) + \lambda E_W(\mathbf{w})$$

Weight Decay (L_2 regularization):

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\phi(\mathbf{x}_n)\right)^2 + \frac{\lambda}{2}\mathbf{w}^T\mathbf{w}$$

Closed-form Solution:

$$\mathbf{w} = (\lambda\mathbf{I} + \mathbf{\Phi}^T\mathbf{\Phi})^{-1}\mathbf{\Phi}^T\mathbf{t}$$

Bayesian Interpretation

Equivalent to MAP estimation with Gaussian prior: $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I})$
where $\lambda = \alpha/\beta$ relates prior precision α to noise precision β

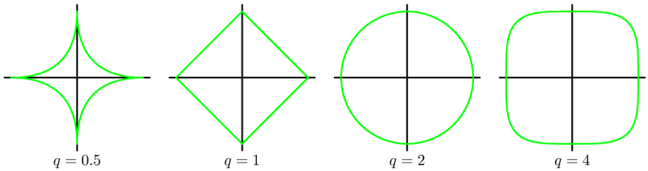
General Regularization

General L_q Regularization:

$$E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^T\phi(\mathbf{x}_n)\right)^2 + \frac{\lambda}{2}\sum_{j=1}^{M}|w_j|^q$$

Common Cases:

- $q = 1$: **Lasso** - sparse solutions
- $q = 2$: **Ridge** - shrink coefficients
- $q < 1$: Non-convex regularization



Sparsity Property

Lasso ($q = 1$) tends to produce sparse models with many zero weights

The Bias-Variance Decomposition

Expected Squared Loss Decomposition:

$$\mathbb{E}[(y(\mathbf{x}) - t)^2] = \underbrace{(\mathbb{E}[y(\mathbf{x})] - \mathbb{E}[t|\mathbf{x}])^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(y(\mathbf{x}) - \mathbb{E}[y(\mathbf{x})])^2]}_{\text{Variance}} + \underbrace{\mathbb{E}[(t - \mathbb{E}[t|\mathbf{x}])^2]}_{\text{Noise}}$$

Intuitive Interpretation:

- **Bias**: Difference between average prediction and true conditional expectation
- **Variance**: Variability of predictions across different datasets
- **Noise**: Irreducible uncertainty in the data

Practical Implications:

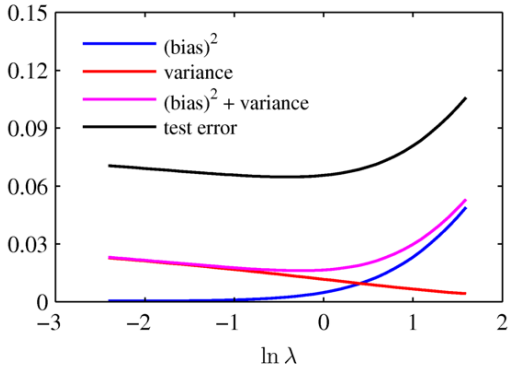
- Simple models: Low variance, High bias
- Complex models: High variance, Low bias
- Goal: Find optimal trade-off

Model Complexity Trade-off

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

Regularization helps control this trade-off by adjusting model complexity

Bias-Variance Trade-off



- **Small λ (Weak regularization)**:
  - Low bias, high variance
  - Overfitting regime
  - Complex model fits noise
- **Large λ (Strong regularization)**:
  - High bias, low variance
  - Underfitting regime
  - Oversimplified model
- **Optimal λ**:
  - Balances bias and variance
  - Minimizes total error

Finding the Sweet Spot

The optimal regularization strength λ^* minimizes the sum of bias^2 and variance

Conjugate Priors: Key Bayesian Concept

Definition: Prior is conjugate to likelihood if posterior belongs to same family.

Bayesian Update:

$$\underbrace{p(\boldsymbol{\theta}|\mathcal{D})}_{\text{Posterior}} \propto \underbrace{p(\mathcal{D}|\boldsymbol{\theta})}_{\text{Likelihood}} \times \underbrace{p(\boldsymbol{\theta})}_{\text{Prior}}$$

Example: Gaussian-Gaussian

- Prior: $\mathcal{N}(\mu_0, \sigma_0^2)$
- Likelihood: $\mathcal{N}(\mu, \sigma^2)$
- Posterior: Gaussian

Advantages:

- Analytical solutions
- Efficient computation
- Interpretable updates

In Bayesian Linear Regression

Gaussian prior + Gaussian likelihood $\rightarrow$ Gaussian posterior

Enables exact closed-form solutions for posterior and predictive distributions

Bayesian Linear Regression

Prior Distribution:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0)$$

Posterior Distribution:

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_N, \mathbf{S}_N)$$

$$\begin{aligned}
\ln p(\mathbf{w}|\mathbf{t}) &= \ln p(\mathbf{t}|\mathbf{w}) + \ln p(\mathbf{w}) + \text{const} \\
&= -\frac{\beta}{2}\sum_{n=1}^{N}(t_n - \mathbf{w}^T\phi_n)^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^T\mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) + \text{const} \\
&= -\frac{1}{2}\mathbf{w}^T(\beta\mathbf{\Phi}^T\mathbf{\Phi} + \mathbf{S}_0^{-1})\mathbf{w} + \mathbf{w}^T(\beta\mathbf{\Phi}^T\mathbf{t} + \mathbf{S}_0^{-1}\mathbf{m}_0) + \text{const}
\end{aligned}$$

$$\ln p(\mathbf{w}|\mathbf{t}) = \ln \mathcal{N}(\mathbf{w}|\mathbf{m}_N, \mathbf{S}_N) = -\frac{1}{2}\mathbf{w}^T\mathbf{S}_N^{-1}\mathbf{w} + \mathbf{w}^T\mathbf{S}_N^{-1}\mathbf{m}_N + \text{const}$$

Posterior Parameters:

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \beta\mathbf{\Phi}^T\mathbf{\Phi}, \quad \mathbf{m}_N = \mathbf{S}_N(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\mathbf{\Phi}^T\mathbf{t})$$

Predictive Distribution

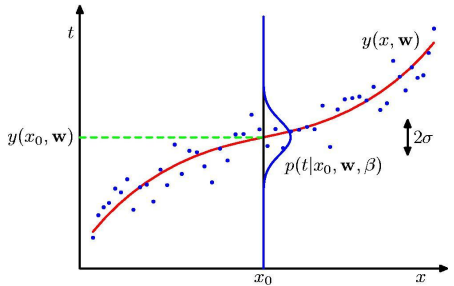
Predictive Distribution:

$$p(t|\mathbf{x}, \mathbf{t}, \alpha, \beta) = \int p(t|\mathbf{x}, \mathbf{w}, \beta) p(\mathbf{w}|\mathbf{t}, \alpha, \beta) d\mathbf{w}$$

Predictive Distribution Form:

$$p(t|\mathbf{x}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(t|\mathbf{m}_N^T \phi(\mathbf{x}), \sigma_N^2(\mathbf{x}))$$

$$\sigma_N^2(\mathbf{x}) = \frac{1}{\beta} + \phi(\mathbf{x})^T \mathbf{S}_N \phi(\mathbf{x})$$



- Blue curve: conditional distribution $p(t|x)$ at a specific input x_0

Practical Uncertainty Quantification

The predictive distribution answers: "Given the data, what is the probability that the target value falls within a certain range at this input location?"

Evidence Approximation: Motivation and Method

The Challenge with Full Bayesian Treatment:

- Ideally, we want to integrate out all parameters:

$$p(t|\mathbf{t}) = \iiint p(t|\mathbf{w},\beta)p(\mathbf{w}|\mathbf{t},\alpha,\beta)p(\alpha,\beta|\mathbf{t})d\mathbf{w}d\alpha d\beta$$

- This triple integral is analytically intractable
- Numerical integration is computationally expensive

Evidence Approximation Solution:

- Approximate $p(\alpha,\beta|\mathbf{t})$ by point estimates $\hat{\alpha},\hat{\beta}$
- Find optimal hyperparameters by maximizing marginal likelihood:

$$\hat{\alpha},\hat{\beta} = \arg\max_{\alpha,\beta} p(\mathbf{t}|\alpha,\beta)$$

- Use these point estimates for prediction:

$$p(t|\mathbf{t}) \approx p(t|\mathbf{t},\hat{\alpha},\hat{\beta}) = \int p(t|\mathbf{w},\hat{\beta})p(\mathbf{w}|\mathbf{t},\hat{\alpha},\hat{\beta})d\mathbf{w}$$

Evidence Maximization: Finding Optimal Hyperparameters

Marginal Likelihood (Evidence):

$$p(\mathbf{t}|\alpha,\beta) = \int p(\mathbf{t}|\mathbf{w},\beta)p(\mathbf{w}|\alpha)d\mathbf{w}$$

Why Maximize Evidence?

The evidence $p(\mathbf{t}|\alpha,\beta)$ measures how likely the data is under the model with given hyperparameters α, β.
Maximizing evidence = Finding hyperparameters that best explain the observed data.

Log Evidence for Linear Regression:

$$\ln p(\mathbf{t}|\alpha,\beta) = \frac{M}{2}\ln\alpha + \frac{N}{2}\ln\beta - E(\mathbf{m}_N) + \frac{1}{2}\ln|\mathbf{S}_N| - \frac{N}{2}\ln(2\pi)$$

where: $\mathbf{S}_N^{-1} = \alpha\mathbf{I} + \beta\mathbf{\Phi}^T\mathbf{\Phi}$, $\mathbf{m}_N = \beta\mathbf{S}_N\mathbf{\Phi}^T\mathbf{t}$, $E(\mathbf{m}_N) = \frac{\beta}{2}\|\mathbf{t} - \mathbf{\Phi}\mathbf{m}_N\|^2 + \frac{\alpha}{2}\mathbf{m}_N^T\mathbf{m}_N$

Fixed Point Equations and Iterative Solution

Optimization Approach:

- Set derivatives to zero: $\frac{\partial}{\partial \alpha} \ln p(\mathbf{t}|\alpha, \beta) = 0$, $\frac{\partial}{\partial \beta} \ln p(\mathbf{t}|\alpha, \beta) = 0$
- Leads to fixed point equations that must be solved iteratively

Fixed Point Equations:

$$\hat{\alpha} = \frac{\gamma}{\mathbf{m}_N^T \mathbf{m}_N}$$

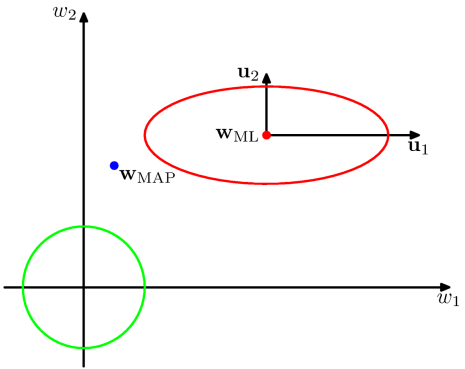
$$\frac{1}{\hat{\beta}} = \frac{1}{N - \gamma} \sum_{n=1}^{N} \left(t_n - \mathbf{m}_N^T \boldsymbol{\phi}(\mathbf{x}_n) \right)^2$$

Iterative Solution Process

These are coupled equations - γ depends on α and β

Solution: Initialize α, β, then iterate until convergence:

Effective Parameters and Model Complexity



Effective Parameters:

$$\gamma = \sum_{i=1}^{M} \frac{\lambda_i}{\alpha + \lambda_i}$$

Key Insight:

- λ_i: Data strength along eigenvector i
- α: Prior strength
- $\lambda_i \gg \alpha$: Parameter well-determined by data
- $\lambda_i \ll \alpha$: Parameter dominated by prior

Evidence-Based Complexity Control

The evidence framework automatically balances data fitting against model complexity by comparing eigenvalues λ_i (data evidence) with prior precision α

Limitations of Fixed Basis Functions

Key Limitation:

- **Curse of dimensionality**: M basis functions along each dimension of a D-dimensional input space requires M^D
- **Rigid structure**: Fixed basis functions cannot adapt to data characteristics

Motivation for Advanced Methods

These limitations motivate more flexible approaches in subsequent chapters

Advanced Approaches in Later Chapters

- Neural networks: Adaptive basis functions (Chapter 5)
- Kernel methods: Implicit feature spaces (Chapter 6)

Summary

- **Linear Basis Models**: Flexible framework for regression with nonlinear basis functions
- **Maximum Likelihood**: Equivalent to least squares under Gaussian noise assumption
- **Bias-Variance Trade-off**: Fundamental concept in model selection
- **Bayesian Approach**: Natural framework for regularization and uncertainty quantification
- **Evidence Approximation**: Empirical Bayes method for hyperparameter estimation
- **Predictive Distributions**: Provide complete uncertainty information for predictions

Key Contribution

Bayesian linear regression provides a principled probabilistic framework that naturally handles regularization and uncertainty quantification, setting the stage for more advanced methods.